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Paleomagnetism of the Triassic red beds of the lower Fundy Group and Mesozoic tectonism of the Nova Scotia platform, Canada

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Abstract

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The lower Fundy Group red beds of the Wolfville and lower Blomidon formations in Nova Scotia and of Wolfville-equivalent units in New Brunswick are exposed around the Bay of Fundy. These red beds were sampled at twenty-six shoreline sites for paleomagnetic analysis, primarily by thermal step demagnetization methods. Samples from four sites proved too fragile to core; however, coating many of the rest of these poorly-indurated samples and cores with waterglass enabled them to withstand the rigours of coring, measurement and thermal cleaning. A stable hematite remanence was found in all twenty-one coherent sites with normal, reversed and mixed polarities. Conglomerate and fold tests indicate the remanence is either primary or very early diagenetic. Its mean direction is $D = 196.5°$, $I = -3.8°$ ($\alpha_{95} = 7.2°$) which converts to a pole position of 45.3° N, 97.1° E ($dp = 3.6°$, $dm = 7.2°$). This pole indicates that the lower Fundy Group was likely deposited in the Early Triassic, rather than in the Middle and Late Triassic, as currently thought. It also indicates that the Nova Scotia platform was rotated about 8° clockwise after deposition of the lower Fundy Group and before extrusion of the North Mountain basalts. The consequences are briefly discussed for two alternative tectonic models of Mesozoic motion on the Minas Geofracture that led to formation of the Bay of Fundy rift as the Atlantic Ocean basin started to open in Early rather than Middle Triassic time. The Fundy Group pole is similar to those from granitic plutons in Maine, thereby increasing the probability that there is a significant error in the mid-Triassic portion of the apparent polar wander path for North America. Finally, magnetostratigraphic methods appear to be a promising way to subdivide and correlate between these red beds.

Introduction

Triassic red beds in the lower part of the Fundy Group outcrop extensively along the Nova Scotia side of the Bay of Fundy, where they are called the Blomidon and Wolfville formations, and in three small areas on the New Brunswick shoreline. These red beds form an interesting paleomagnetic target for several reasons. (1) They are among the youngest and least metamorphosed red beds in Canada; yet the stability of their remanence can be checked using conglomerate, fold and igneous contact tests which might provide information on how and when red beds are magnetized. (2) They are not very amenable to either radiometric or paleontologic dating so that their age within the Permo-Triassic is not well established. Paleomagnetism might provide more precise dating. (3) They were the first units to be deposited as rifting began between North America and Europe to form the Atlantic Ocean and they might, therefore, well provide useful geotectonic information relating to the development of the Bay of Fundy rift. Despite these obvious attractions the Triassic

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red beds had not been studied paleomagnetically. The reason is likely because they are very poorly indurated so that it is difficult to prepare coherent specimens for measurement.

Geology

This synopsis of the geology of the Fundy Group has been drawn mainly from the descriptions and discussions of Klein (1962), Hubert and Hyde (1982), Hubert and Mertz (1984) and Nadon and Middleton (1985). The group outcrops extensively along the Nova Scotia side of the Bay of Fundy and in a few small areas along the New Brunswick side (Fig. 1). It also underlies most of the Bay of Fundy and the Gulf of Maine (Ferguson and Fyffe, 1985).

The Fundy Group rests unconformably on the upper Pennsylvanian Pictou Group and on older units of the Meguma Belt in Nova Scotia and abuts in fault contact with the Avalon Belt in New Brunswick (Keppie, 1979; Bujak and Donahue, 1980). In the Hercynian transitional zone during Permian to Early Triassic time this area was an uplifted part of Pangea with a hot, arid climate. The breakup of Pangea in the Triassic began as rifting and led to the formation of the Fundy Basin into which the continental red beds and tholeiitic basalts of the Fundy Group were deposited. Compositionally, the red beds include graywacke, arkose and orthoquartzites.

The red beds of the basal Wolfville Formation of the Fundy Group are medium to coarse-grained, thick-bedded to massive, red to brown sandstones with interbedded "sharpstone" conglomerates. These beds are up to 1800 m thick in Nova Scotia. The 1800 m thick Lepreau Formation at Point Lepreau and the 1300 m thick Honeycomb Point and Quaco formations at St. Martin's in New Brunswick are stratigraphically equivalent to the Wolfville, as are the red beds at Waterside. These basal red beds were deposited on the margins of the Fundy Basin as interbedded alluvial fans and fluvial channels on a flood plain. The sediments were derived from the surrounding highlands in Nova Scotia and New Brunswick.

The red beds of the overlying Blomidon Formation are fine to medium-grained, laminated to thin-bedded, pale red-brown to olive-gray shales with interbedded eolian sandstones that are up to 400 m thick in Nova Scotia. The 1300 m thick Echo Cove Formation at St. Martin's is stratigraphically equivalent. These lacustrine red beds were deposited in the interior of the basin as sandflats, playa mudflats and playa lakes under shallow semitropical conditions both in the active wave zone and below it.

The 300 m thick tholeiitic North Mountain basalts overlie the Blomidon. Their extrusion in

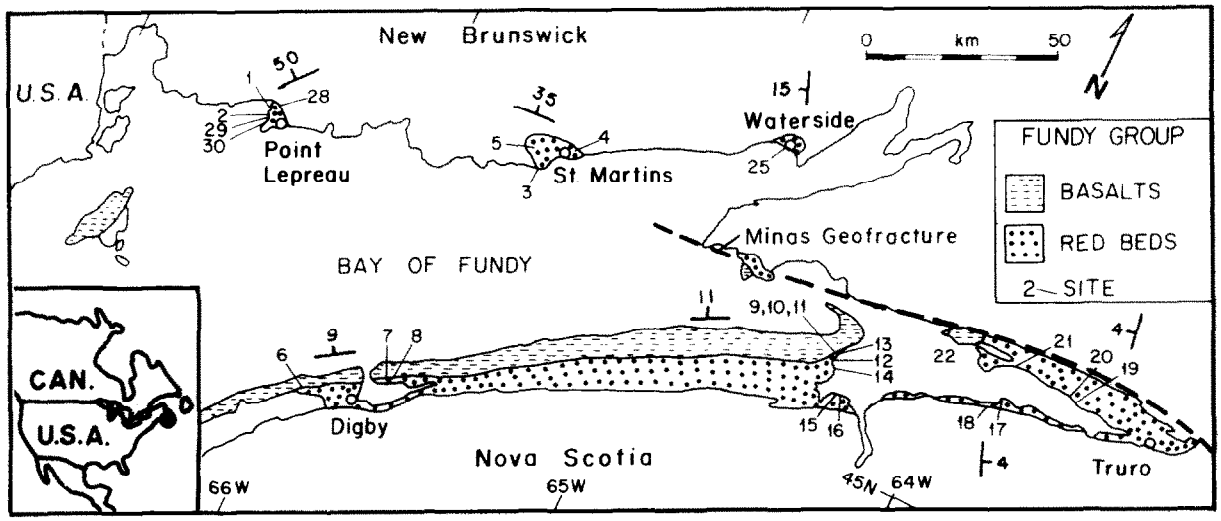


Fig. 1. Location map for the sampling sites in the Fundy Group.

the Early Jurassic marks the onset of rift volcanism with the active separation of North America from Europe and Africa to form the Atlantic Ocean. A thin 10 m unit of red beds over the basalts is called the Scots Bay Formation and is the uppermost formation of the Fundy Group. Neither the North Mountain basalts nor the Scots Bay red beds were sampled in this study.

The age of the Wolfville, Blomidon and equivalent red beds is uncertain. They unconformably overlie the upper Pennsylvanian Pictou Group and the Lepreau Formation contains Pictou basalt clasts. This sets an older age limit of about 290 Ma. The younger age limit is set in the Early Jurassic by the K–Ar whole isochron age of 191 ± 2 Ma determined for the North Mountain basalts by Hayatsu (1979). Within this 100 Ma timespan there is little evidence of the exact age of the Lower Fundy Group. The Wolfville contains reptilian remains that indicate an upper Middle (Ladinian) to lower Late Triassic (Carnian) age, and the Echo Cove Formation contains plant fossils, reptile tracks, spores and pollens that indicate a late Late Triassic age. Note, however, that deposition of the Wolfville could well have begun in Early Permian time, based on the evidence.

The Fundy Basin subsided on high-angle normal faults to form a down-faulted half-graben. The Fundy Group dips gently ($\leq 10°$) into the rift from the Nova Scotia side and much more steeply (up to 70°) from the New Brunswick side. The Cobequid–Chedabucto fault zone or Minas Geofracture (Keppie, 1982) is a left-lateral system that separates the Meguma Belt of the Nova Scotia side from the Avalon Belt or New Brunswick side of the basin.

Metamorphism in the Fundy Group is minimal, reaching zeolite facies rank at most (Keppie and Muecke, 1979).

Experimental methods

Sampling

Three or four oriented block samples were collected at each of twenty-five sites in the Wolfville, Lepreau, Honeycomb Point, and Blomidon Formations (Fig. 1, Table 1). At one additional site (No. 28) in the Lepreau Formation, seventeen basalt clasts were collected to provide a conglomerate test.

Specimen preparation

Many of the block samples, particularly from the Blomidon shales, were very fragile. Samples from four sites (Nos. 7, 10, 14, 21) disintegrated prior to coring. About half of the blocks were poorly indurated and were coated with waterglass. Waterglass is sodium metasilicate. It comes as a powder called "Metsil" that is mixed with water to form a gel or slurry which penetrates into the surface pores of the rock to provide a glassy binding agent that can withstand 700 °C temperatures. Cores were drilled from each block using a Felker thin-wall bit with minimal water in order to prevent washing. After coating the many fragile cores with Metsil, the cores were sliced to produce three or four specimens per block. The weak specimens were again coated with Metsil, after which they proved sufficiently coherent to withstand the subsequent measurement and thermal demagnetization treatments.

NRM

The natural remanent magnetization (NRM) of each specimen was measured using an automated two-axis CTF cryogenic magnetometer inside a shielded room with an ambient magnetic field of about 0.1 nT. The NRM directions are widely scattered with some bias towards the present Earth's magnetic field (PEMF) direction and towards the north and south horizontal. The NRM intensities are mostly in the upper 10^{-4} to the lower 10^{-3} A m^{-1} range.

AF step demagnetization

Two specimens from each site were alternating field (AF) demagnetized in nine steps from 5 mT up to 100 mT using a Schonstedt GSD-1 AF Demagnetizer. A majority of red bed specimens showed short demagnetization tracks away from the PEMF of $D = 338°$, $I = 73°$ (declination, inclination) without reaching a stable end point

TABLE 1

Site collection and mean remanence data *

Site No.	Unit Fm.	Attitude		Cleaning (°C)	Mean Remanence Direction					
		strike (°)	dip (°)		N	decl. (°)	incl. (°)	α_{95} (°)	k	pol
1	Lep	244	32	540–640	13	221	−16	17	7.2	S
2	Lep	215	53	540–640	11	220	−20	12	15.0	M
3	Hon	261	41	540–640	12	189	−35	21	5.3	M
4	Hon	240	28	540–640	11	204	−11	20	6.3	S
5	Hon	290	34	540–660	7	190	−24	21	9.1	S
6	Blo	230	6	620–680	4	200	−29	9	107.0	N
8	Blo	230	9	640–680	6	194	−16	12	34.5	N
9	Blo	240	11	620–680	9	196	−15	11	23.3	M
11	Blo	240	11	620–680	6	192	−27	23	9.2	S
12	Blo	240	11	620–640	14	202	−7	11	12.4	M
13	Blo	240	11	620–680	12	191	8	17	7.7	S
15	Wol	70	11	620–680	13	182	5	19	5.6	S
16	Wol	70	11	620–680	10	167	13	15	11.5	S
17	Wol	355	4	620–680	9	194	14	22	6.4	M
18	Wol	330	4	620–680	9	198	−23	19	8.3	N
19	Wol	170	3	620–680	11	212	−6	17	8.1	M
20	Wol	140	4	620–680	13	216	−11	15	8.8	M
22	Blo	222	31	620–640	13	188	−9	6	42.0	N
25	Wol	155	15	540–640	8	181	0	8	46.4	N
28	LCg	218	26	500–590	17	102	63		1.6	
29	Lep	212	72	540–640	10	192	−31	27	4.1	N
30	Lep	200	60	540–660	8	194	−37	23	7.0	N

* The "Unit Formation" abbreviations are: Blo = Blomidon; Hon = Honeycomb Point; LCg = Lepreau conglomerate clasts; Lep = Lepreau; and Wol = Wolfville. The "Attitude" is given by the strike and dip in degrees using a right hand convention. The "Cleaning" indicates the effective thermal demagnetization treatment in degrees Celsius. The "Mean Remanence Direction" is calculated with all northward vectors reported to their antiparallel southward position before tectonic correction and it is defined by: the number of specimens (N); the declination, inclination and radius of the cone of 95% confidence (Decl., Incl., α_{95}) in degrees; the precision parameter (k) of Fisher (1953); and the polarity of the specimens to the north (N), south (S), or mixed north and south (M).

(Fig. 2a). A minority of specimens apparently do reach a stable direction but show little intensity decay on so doing (Fig. 2b). A very few specimens from Point Lepreau do show some progressive decay of a stable component which, on tectonic correction, align in the north population. This is likely because the recent viscous remanent magnetization (VRM) component acquired in the PEMF aligns with the tilted characteristic remanence component (Fig. 2c). The basalt clast specimens do have a significant linear decay segment that is indicative of a stable component residing in domains with AF activities that are typical of magnetite (Fig. 2d). In general, however, AF demagnetization was not as effective as thermal demagnetization and was not used any further.

Thermal step demagnetization

Another two specimens from each site were thermally demagnetized in eleven steps up to 680 °C using a Schonstedt TSD-1 Demagnetizer. After tectonic correction for bedding tilt, nearly all of the specimens behaved similarly. Those specimens with NRM directions that were horizontal and to the north or south showed the removal of VRM components up to about 500 °C, slow linear decay between 500 °C and 660 °C, and

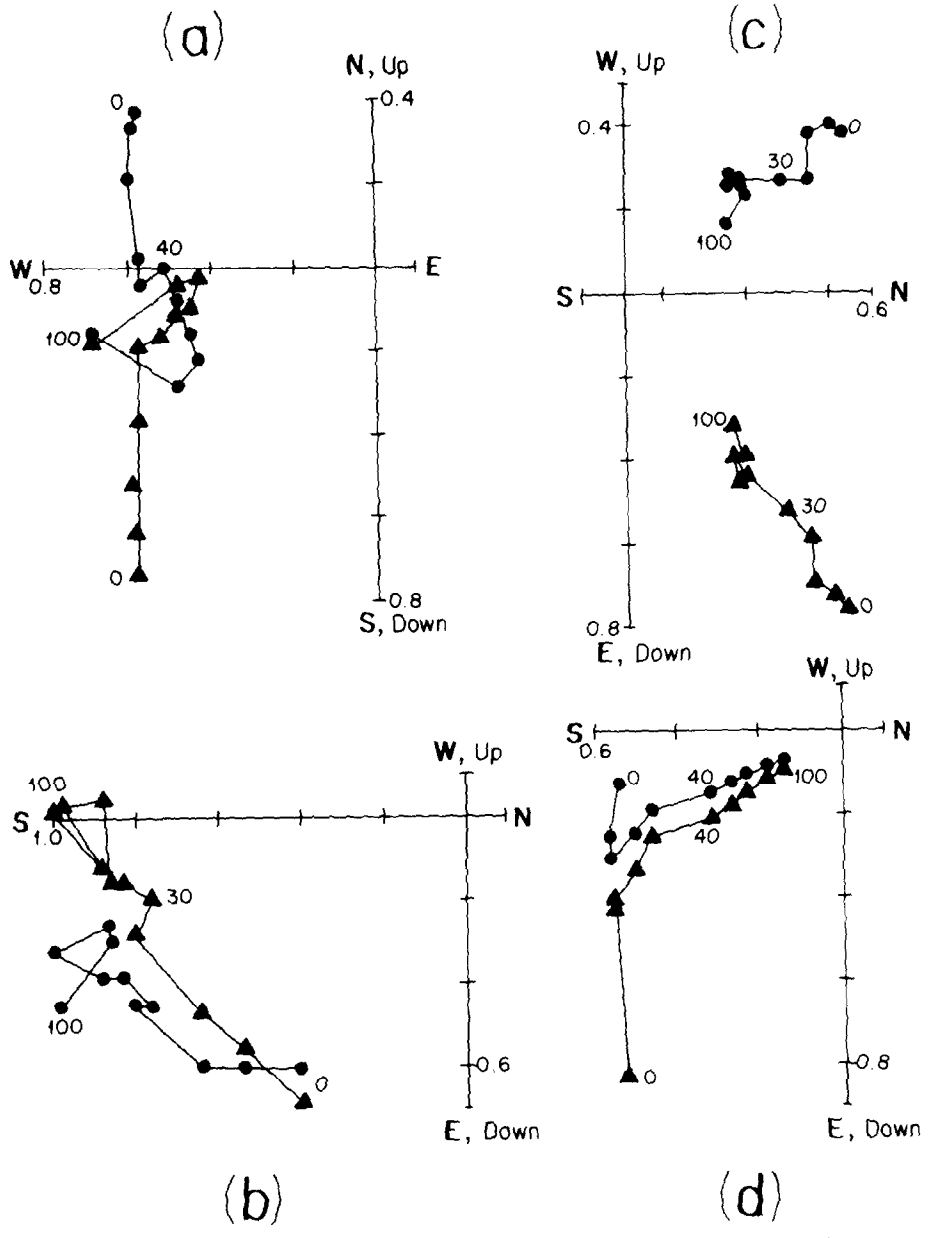


Fig. 2. Orthogonal vector decay plots of AF step demagnetization data for example specimens from: (a) site 2; (b) site 5; (c) site 30; and (d) site 28 (basalt clast). The horizontal and vertical plane projections are shown by circles and triangles respectively. The axial lengths are expressed as a ratio of the NRM intensity. The point values are in milliTesla (mT).

rapid decay between 660 °C and 680 °C as the Néel temperature of hematite is reached (Fig. 3a, b). These stable hematite directions are horizontal and to the north or south.

Those specimens with NRM directions close to the PEMF direction typically show a rapid swing away from that direction toward low inclinations up to an unblocking temperature of about 500 °C, a continued slow swing up to about 620 °C, and then a linear decay to the origin between 620 °C and 680 °C as the same horizontal and north or south stable end points are defined (Fig. 3c).

The remaining red bed specimens show an intermediate behaviour with the removal of a moderate VRM component up to 500 °C, some further decay up to 640 °C, and then a fairly rapid decay up to 660 °C or 680 °C to define horizontal southward and northward end points (Fig. 3d).

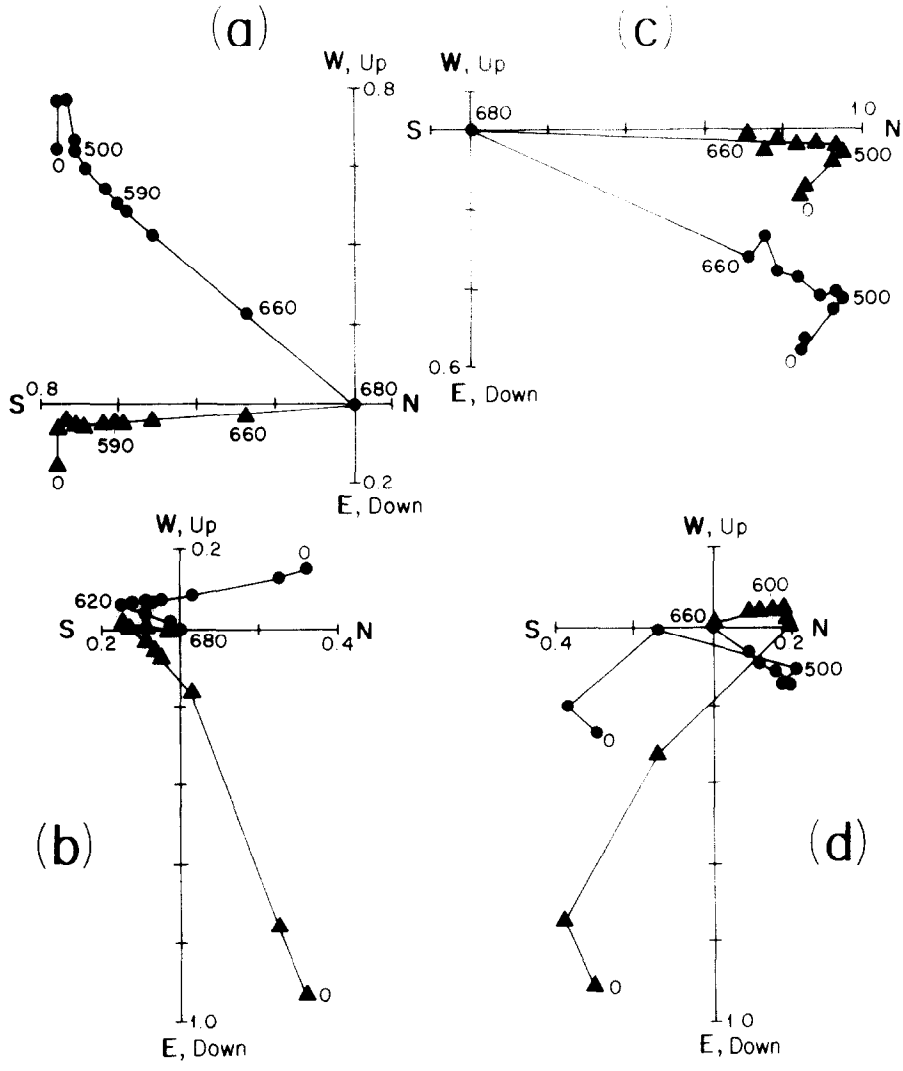


Fig. 3. Orthogonal vector decay plots of thermal step demagnetization data for example red bed specimens from: (a) site 4; (b) site 22; (c) site 9; and (d) site 13. Conventions as in Fig. 2, except that point values are in degrees Celsius.

As the 675 °C Néel temperature of hematite in the red beds was reached, a few specimens showed polarity switches between the north and south horizontal of the sort described by Roy and Park (1974).

All of the red bed specimens that had not been thermally step demagnetized in detail, including those that had been AF step demagnetized, were thermally step demagnetized at 200 °C, 640 °C, 660 °C, and 680 °C. Their characteristic remanence directions were obtained by averaging the two or more consistent directions with significant intensity decay. Except for a few specimens with remanence intensities close to the 1×10^{-5} A m^{-1} noise level of the magnetometer, stable directions were found. When polarities are examined, the specimens from seven sites are north and horizontal, from seven sites are south and horizontal, and from seven sites contain both north and south and horizontal directions. This indicates that the magnetization was acquired during a period in which the Earth's magnetic field underwent several polarity reversals.

After changing the northward directions to their

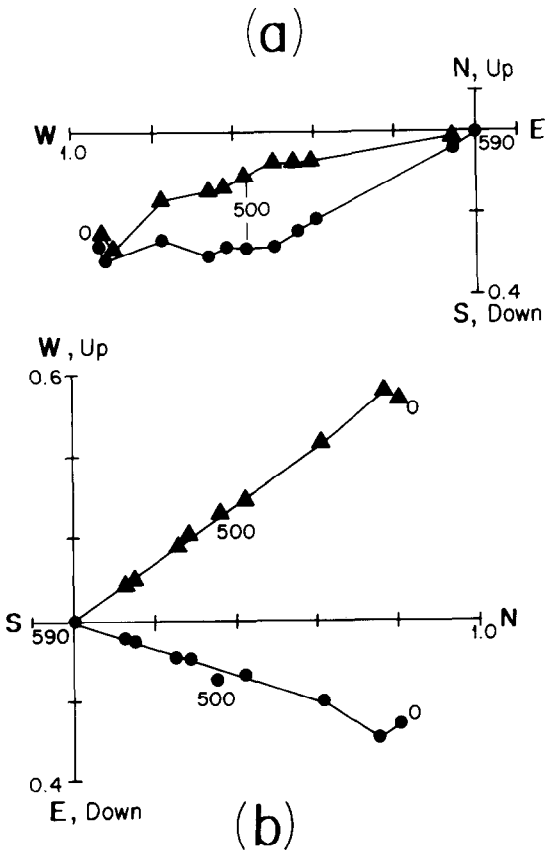


Fig. 4. Orthogonal vector decay plots of thermal step demagnetization data for example basalt clast specimens from site 28. Conventions as in Figure 3.

antiparallel southward positions, the specimen directions were averaged to obtain the site mean remanence directions (Table 1).

The basalt clasts show stable linear decay on thermal demagnetization with very small initial VRM components. The remanence is well-defined between 500 °C and 590 °C as the Curie temperature of magnetite is exceeded (Fig. 4a, b). It is evident that the magnetization resides in very stable magnetite domains and that the clasts have not been significantly altered by hematization since deposition.

There is no apparent difference in the directions obtained from specimens coated with Metsil and those that did not require coating.

Analysis and discussion

Conglomerate test

Two specimens were analyzed for each of eight Pennsylvanian basalt clasts from the Lepreau conglomerate, and they give an average interspecimen angular difference for their stable remanence of 25° ± 18°. This indicates that the clast remanence

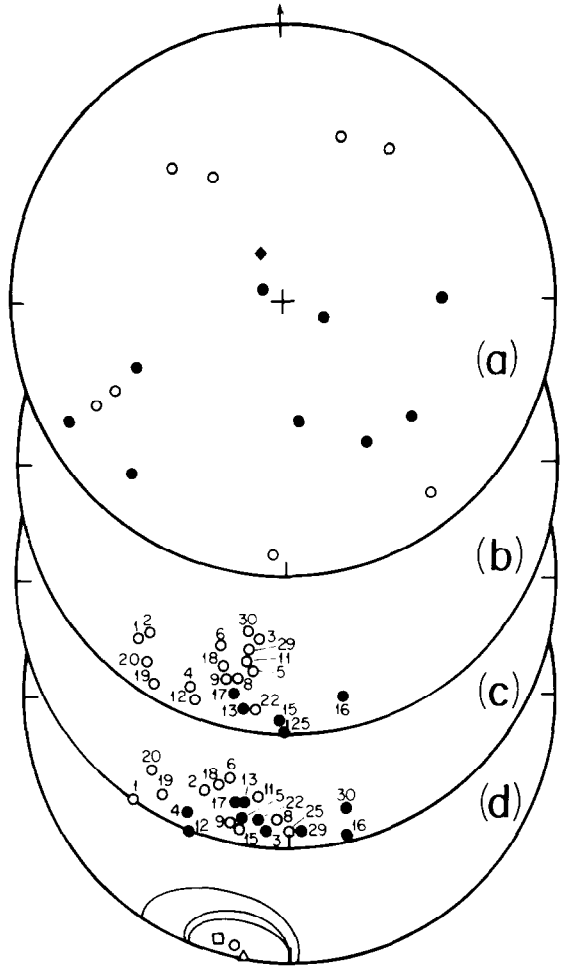


Fig. 5. Equal-area stereonet plots showing remanence directions for: (a) basalt conglomerate clasts of site 28; (b) site means for red beds uncorrected; (c) site means for red beds after tilt correction; and (d) average of corrected site means for the Wolfville Formation of Nova Scotia (square), for the Wolfville equivalents of New Brunswick (triangle), and for the Blomidon Formation of Nova Scotia (circle) with their cones of 95% confidence (Fisher, 1953). Solid and open symbols denote downwardly and upwardly directed vectors respectively. The diamond shows the present Earth's magnetic field direction.

TABLE 2

Fold test data *

Group	N	R	Decl. (°)	Incl. (°)	k
1. Flat-lying Sites —uncorrected	6	5.5894	194.9	1.4	12.18
2. —tilt corrected	6	5.5894	196.3	−5.2	12.18
3. Inclined Sites—uncorrected	6	5.8036	201.7	−27.9	25.45
4. —tilt corrected	6	5.8036	189.6	−1.4	25.45
5. Combined Sites —uncorrected	12	11.0711	198.1	−14.9	11.84
6. —tilt corrected	12	11.3212	192.9	−3.2	16.21

* *Notes*

(A) $K_3/K_1 = 2.09 < F_{95\%\ confidence} = 2.98$ and therefore both limbs have comparable precisions

(B) $(R_1 + R_3 - R_5^2)/(R_1 + R_3))/2(N - R_1 - R_3) = 0.523$ is > 0.349 and therefore the uncorrected limbs have different mean directions

(C) $(R_1 + R_3 - R_6^2)/(R_1 + R_3))/2\ (N - R_1 - R_3) = 0.040$ is < 0.349 and therefore the tilt corrected limbs have the same mean direction at the 95% confidence level.

(D) The conventions are as given in Table 1 except R is the vector resultant of Fisher (1953).

is reasonably stable and internally consistent. When all seventeen clast directions are plotted, they appear to be quite random (Fig. 5a). When they are averaged, they give a resultant vector (R) of 7.04 (Fisher, 1953) which is slightly more than the minimum 95% confidence value of 6.60 for non-random directions (Watson, 1956a). Thus the conglomerate test (Graham, 1949) states with 93% confidence that the clast directions are random, and therefore that the clasts have not been metamorphosed or altered to give a common direction since deposition. This also implies that the remanence of the red beds is primary, i.e. it cannot have been produced by a secondary thermal metamorphic event and any secondary chemical event would have had to alter completely the entire lower Fundy Group except for the clasts, which is unlikely.

Fold test

A cursory examination of the site mean directions without tilt correction (Fig. 5b) indicates a tectonic bias. Following McFadden and Jones (1981), the six flat-lying sites (Nos. 15–20) with dips of $< 12°$ from the Wolfville Formation in Nova Scotia were compared in a fold test to the six inclined sites (Nos, 1–3, 5, 29 and 30) with dips of $> 30°$ from the equivalent units in New Brunswick (Table 2). The test shows at the 95% confidence level that the two limbs have significantly different mean directions before tectonic

TABLE 3

Group mean remanence directions *

Group	Sites	Mean remanence direction				
		N	decl. (°)	incl. (°)	α_{95} (°)	k
Wolfville Fm., N.S. **	15–20	6	196.3	−5.2	18.0	14.7
Wolfville Eq., N.B. **	1–5, 25, 29–30	8	190.2	−1.1	13.0	19.0
Blomidon Fm., N.S. **	6–13, 22	7	191.9	−5.7	12.2	25.4
All-uncorrected	all above	21	196.2	−13.5	7.9	17.1
All-tilt corrected **	all above	21	192.5	−3.8	7.2	20.5

* Conventions as in Table 1.

** denotes use of tilt-corrected data.

correction (i.e. 0.523 > 0.349 in Table 2), whereas they do not after tectonic correction (i.e. 0.040 < 0.349 in Table 2). Thus the fold test is positive and concludes that the lower Fundy Group red beds acquired their magnetization before the folding occurred. The age of folding in the Bay of Fundy is Triassic to Early Jurassic, coincident with faulting and the infilling of the basin with about 12 km of Fundy Group sediments (Ferguson and Fyffe, 1985).

Test for stratigraphic or geographic bias

As a check for internal differences within the population of site means from stratigraphic or regional geographic causes, the means were subdivided into three subsets: (1) the Wolfville sites in Nova Scotia; (2) the Wolfville-equivalent sites in New Brunswick; and (3) the Blomidon sites in Nova Scotia. All three subsets give similar mean directions that fall within the cones of 95% confidence of each other, indicating that all three are statistically the same (Fig. 5d, Table 3). Given the small size of each subset, it is not surprising that the cones of 95% confidence are relatively large. While more sampling might establish significant differences, there is no reason at present to subdivide the population on a stratigraphic or geographic basis.

Magnetostratigraphy

The lower Fundy Group contains no marker horizons of regional extent at present. Magnetostratigraphy looks like a promising tool (Fig. 6). The stratigraphically lowest sites near the base of the Wolfville and Lepreau formations on either side of the Bay of Fundy are normal (southward) in polarity with mixed polarity then reversed polarity sites above. The Honeycomb Point sites have reversed sites then a mixed polarity site above, and this pattern appears consistent relative to the upper Lepreau and Wolfville red beds. Also all three of the uppermost Blomidon Formation sites, which are geographically well separated, are normal in polarity. Therefore, although the sites were not collected for magnetostratigraphic purposes, they do provide clear evidence that this technique could provide valuable marker horizons for the stratigraphic analysis of these red beds.

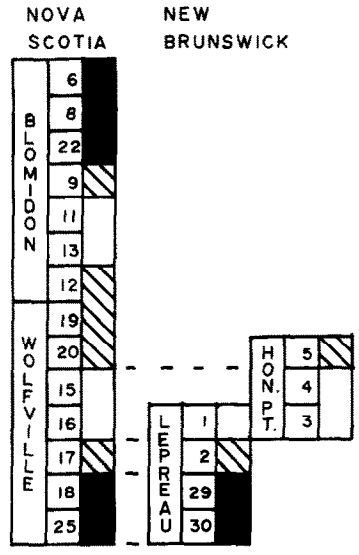


Fig. 6. Initial apparent magnetostratigraphy for the lower Fundy Group red beds. *Hon. Pt.* is the Honeycomb Point Formation. Normal (southward) polarity sites are solid blocks, mixed polarity sites are diagonally-lined blocks and reversed (northward) polarity sites are open blocks.

Pole positions

The unit mean pole position of the lower Fundy Group after tilt correction is 97.1° E, 45.3° N ($dp = 3.6°$, $dm = 7.2°$) (longitude, latitude, semi-axes of the oval of 95% confidence). This pole position falls closer to a much older portion of the apparent polar wander path (APWP) for the North American craton (Irving and Irving, 1982) than the poles determined for the overlying North Mountain basalts of the Fundy Group by Carmichael and Palmer (1968) and by Larochelle (1967) (Fig. 7a). The basalts retain only northward directions and give a pole position after tilt correction that is reasonably consistent with their K–Ar radiometric age of 191 ± 2 Ma (Hayatsu, 1979). Carmichael and Palmer (1968) also report a positive conglomerate test using North Mountain basalt clasts in the overlying Scots Bay Formation that indicates that the basalts retain a primary remanence. Thus their data support the results of the demagnetization analyses, of the conglomerate test and of the fold test done in this study that

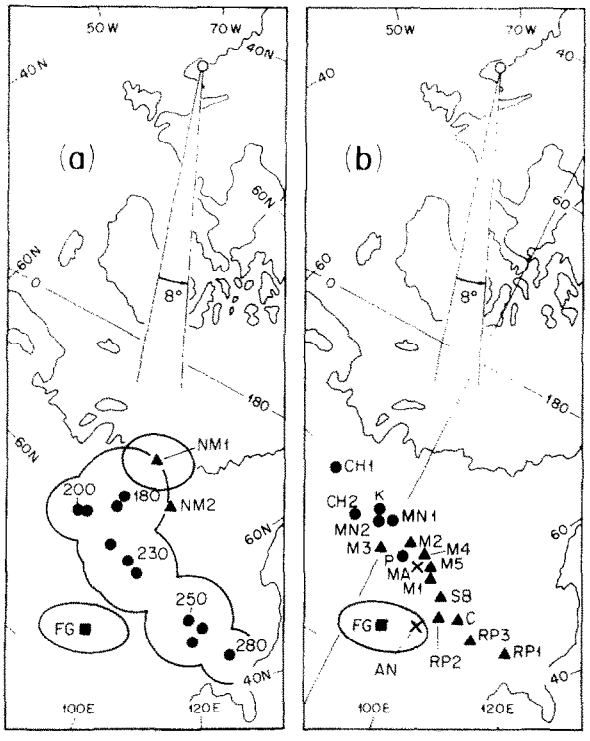


Fig. 7. Pole positions plotted on a segment of the northern hemisphere. *FG* is the lower Fundy Group pole from this study with its oval of 95% confidence. The open circle is the study area with the indicated 8° rotation away from the apparent polar wander path (APWP). Also shown: (a) the APWP from Irving and Irving (1982) with poles averaged for 10 Ma increments from 180 to 280 Ma using a 30 Ma smoothing window with the 95% confidence limits, and the poles from the North Mountain basalts of the upper Fundy Group as triangles: *NM1*—from Carmichael and Palmer (1968), and *NM2*—from Larochelle (1967); (b) other Triassic poles from the North American craton as follows. Late Triassic (circles): Chinle (*CH1*—Utah; *CH2*—N. Mex.), Kayenta (*K*—Utah); Manicouagan (*MN*1, *MN2*—Prov. of Quebec), and Popo Agie (*P*—Wyo.). Middle Triassic (crosses): upper Maroon (*MA*—Colo.) and Ankarah (*AN*—Wyo.). Early Triassic (triangles): Moenkopi (*M1*—Utah; *M2*—Ariz.; *M3*—Utah; *M4* and *M5*—Colo.), State Bridge (*SB*—Colo.), Chugwater (*C*—Wyo.), and Red Peak (*RP1, RP2, RP3*—Wyo.), as listed in Gordon (1984) and Irving and Irving (1982).

show that the lower Fundy Group red beds carry a primary or very early diagenetic remanence.

A significant point is that there is no evidence of modern hematite remanence despite the fact that all of the red bed samples came from the intertidal zone where they have been repeatedly saturated with oceanic water. Nor is there any evidence of a secondary remanence of thermochemical origin related to the extrusion of the North Mountain basalts that have a very different mean direction of $D = 3.6°$, $I = 47.3°$ ($\alpha_{95} = 5°$) with only northwardly directed vectors.

Tectonics

The explanation of the discordant pole position for the lower Fundy Group is not straightforward. Two alternatives appear untenable.

Accepting the conventional age assignment of the Wolfville and Blomidon, the sites would represent an average age of about 225 Ma. Simple translation could move the lower Fundy Group pole (*FG*, Fig. of 7a) to this point on the APWP. In this case, the terrain would have to have been formed about 12° or about 1200 km due south of its present location relative to the craton. This would be beyond the continental slope, not far north of Bermuda, which is inconceivable.

A second possibility would be to argue that these units are about 15 Ma older. In this case, the terrain could have been moved northeastward by sinistral strike-slip motion along the Atlantic seaboard from somewhere near the Carolinas. This would shift the FG pole to coincide with the 240 Ma pole on the APWP. However, no such major post -Permian fault is known and the minor displacements in the Bay of Fundy rift were in the opposite direction (Keppie, 1982).

There are two plausible alternatives. The first invokes a regional rotation of about 8° clockwise, but it also requires that the red beds be significantly older than previously postulated. Correction for an 8° clockwise rotation of the terrain shifts the FG pole to between poles RP3 and *C* in Fig 7b, so that its oval of 95% confidence encircles the lower Early Triassic poles and encroaches on the region of known Late Permian poles. The remanence cannot be much older otherwise it would fall in Kiaman (Late Carboniferous and Permian) time when only southwardly directed remanences would be found rather than the mixed polarities found in these red beds. This implies deposition between about 250 and 240 Ma. Such an age is somewhat older than the scant paleontologic evidence suggests, but is otherwise acceptable. Ziegler (in press) in his model for the evolu-

tion of the Atlantic margin postulates significant sedimentation and rifting in the Bay of Fundy basin between Late Permian and Ladinian time which is consistent with an Early Triassic age. This age also implies either slow deposition of the upper Blomidon or a time break before extrusion of the overlying North Mountain basalts in the Early Jurassic period.

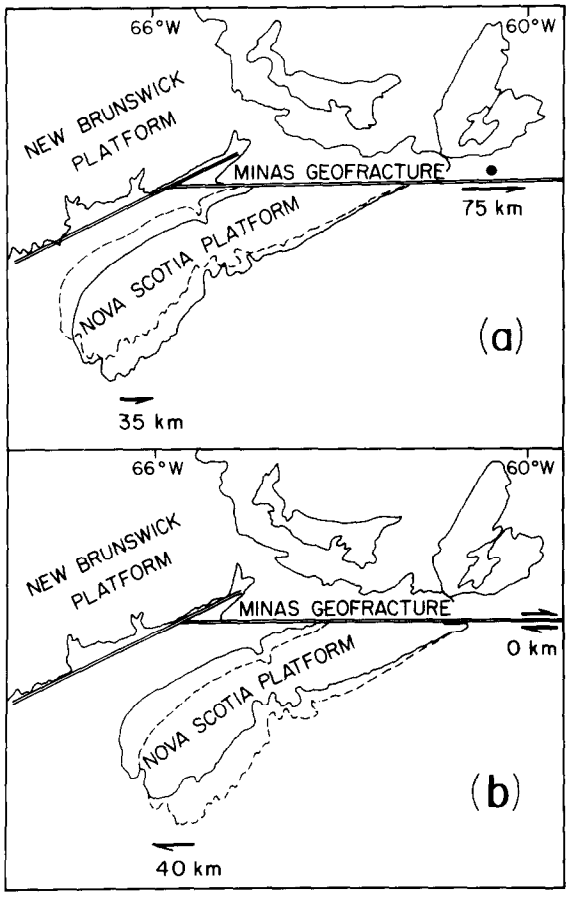


Fig. 8. Tectonic models based on holding the New Brunswick platform and Nova Scotia north of the Minas Geofracture fixed and moving only the Nova Scotia platform. a. The dashed line shows the Permian position for the Nova Scotia platform according to Keppie (1982) prior to 75 km of eastward translation in the Mesozoic. The solid line shows the modified position required by the paleomagnetic data with north end moving 75 km while the south end moves by only 35 km to reach its present location. b. Assuming only dextral movement on the Minas Geofracture (Mawer and White, 1987), the dashed line shows the position of Nova Scotia during deposition of the lower Fundy Group as the platform initially impacts at its north end, and the solid line shows the present position of Nova Scotia after the south end has moved 40 km farther westward to achieve the 8° rotation seen in the paleomagnetic data.

Keppie (1982) accounts for the formation of the Bay of Fundy rift system by translating the Nova Scotia platform eastward by 75 km along the Minas Geofracture in Mesozoic time (Fig. 8a). The requisite 8° clockwise rotation would result if the south end of the Nova Scotia platform moved eastward by only about 35 km during the same time. The repositioning of Nova Scotia at the start of rifting does align the Nova Scotia and New Brunswick margins of the Bay of Fundy rift somewhat better.

Mawer and White (1987) find no evidence of sinistral motion on the Minas Geofracture as Keppie's (1982) model requires. To satisfy the required 8° of clockwise rotation required by the paleomagnetic data with only dextral motion on the Minas Geofracture, the south end of Nova Scotia platform has to move about 40 km farther westward after the north end first abuts and stops against the old Avalon terrain in northern Nova Scotia (Fig. 8b). This model could explain the progressively deeper downwarp of the Fundy basin towards the south. Thus either the sinistral or dextral model can explain tenably the relatively minor 8° clockwise rotation. Much more evidence is required to truly assess these models including a better geologic understanding of the Bay of Fundy rift and a much better paleomagnetic data base for the Nova Scotia platform.

The second plausible alternative is less certain. Recently Wu and Van der Voo (1988) reported Middle to Late Triassic paleopoles of 48° N, 92° E ($\alpha_{95} = 4°$) and 48° N, 99° E ($\alpha_{95} = 3°$) for the small 221 ± 8 Ma Abbott and 228 ± 5 Ma Agamenticus granitic plutons in southern Maine. These poles also fall off the APWP and straddle the Fundy Group pole. Wu and Van der Voo (1988) cite reasons to believe that postintrusive tilt has not affected these plutons or their poles, but cannot entirely discount the possibility in arguing for a "small but significant revision" to the APWP. They suggest that the APWP tracks through the Abbott and Agamenticus poles at 225 Ma which, if true, means that the Fundy Group pole would be concordant with its upper Middle to lower Late Triassic paleontologic age. This in turn would

obviate the need for an Early Triassic age of remanence acquisition or for an 8° clockwise tectonic rotation.

Conclusions

Four conclusions are drawn from this study:

(1) The lower Fundy Group red beds retains a stable primary or early diagenetic remanence in hematite which gives a mean direction of $D = 192.5°$, $I = -3.8°$ ($\alpha_{95} = 7.2°$) after tilt correction.

(2) The Wolfville and lower Blomidon formations appear to be Early Triassic in age rather than Middle to Late Triassic as heretofore thought. This provides evidence for the Early Triassic initiation of rifting in the formation of the Atlantic Ocean basin, coincident with early events in northwestern Europe, Greenland and Arctic Canada (Ziegler, in press).

(3) The Nova Scotia platform appears to have been rotated by about 8° clockwise since the lower Fundy red beds were deposited, presumably in the Early Jurassic period during the initial stages of the opening of the Atlantic Ocean. The possibility remains that the Middle Triassic portion of the North American APWP is mislocated and that the Fundy Group pole is in fact concordant. This would negate the second and third conclusions that are given above.

(4) Magnetostratigraphic methods appear to be promising for use in studying the stratigraphy of these red beds.

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